



EQUIPMENT DATASHEETS & CURVE - SULFUR FEED PUMP

END USER: LITHIUM NEVADA LLC

PO NO.: 4550001187

TAG NO.: 1510-PP-004/5

SERIAL NO.: 77068-1/2

TYPE: VSHR-1682

PROJECT: THACKER PASS PROJECT

PROJECT NO.: HOU-23001048-A1

EQUIPMENT: SULPHUR FEED PUMP

DOC NO.: 1500-PE-VEN-4165-EXP-0902

VDRL CODE: 2009 + 2011 + 2504

VENDOR DATA STATUS

☐ 1 : APPROVED

☐ 2 : APPROVED WITH COMMENTS (RESUBMIT AS FINAL)

☐ 3 : NOT APPROVED (REVISE AND RESUBMIT)

☐ 4 : APPROVAL NOT REQUIRED (FOR INFORMATION ONLY)

REVIEW DATE:

DEPT SIGNATURE:

SUBMISSION STATUS

4	8/20/2025	ISSUE FOR APPROVAL	Mason Seidel WEIR LEWIS PUMPS	Reid Chickering WEIR LEWIS PUMPS
3	7/22/2025	ISSUE FOR APPROVAL	Mason Seidel WEIR LEWIS PUMPS	Reid Chickering WEIR LEWIS PUMPS
2	6/10/2025	ISSUE FOR APPROVAL	Mason Seidel WEIR LEWIS PUMPS	Reid Chickering WEIR LEWIS PUMPS
1	5/19/2025	ISSUE FOR APPROVAL	Mason Seidel WEIR LEWIS PUMPS	Reid Chickering WEIR LEWIS PUMPS
REV	REV DATE	DESCRIPTION	PREPARED BY	REVIEWED BY

Tag number	: 77068-1/2.REV1	Size	: 2VSHR
Pump Type	: VSHR-1682	Stages	: 1
Service	: Molten Sulphur	Based on curve number	: 2-2643-62E
Quantity	: 2	Date last saved	: 22 Jul 2025 2:17 PM
Quote number	: 3000280903		

Operating Conditions

Flow, rated	: 99.36 USgpm
Differential head / pressure, rated (requested)	: 279.1 ft
Differential head / pressure, rated (actual)	: 287.8 ft
Suction pressure, rated / max	: 0.00 / 0.00 psi.g
NPSH available, rated	: Ample
Site Supply Frequency	: 60 Hz

Performance

Speed criteria	: Synchronous
Speed, rated	: 3500 rpm
Impeller diameter, rated	: 8.88 in
Impeller diameter, maximum	: 10.00 in
Impeller diameter, minimum	: 6.00 in
Actual bypass flow (shutoff / rated)	: 0.00 / 0.00 USgpm
Efficiency (bowl / pump)	: 46.23 / 45.67 %
NPSH required / margin required	: 12.18 / 0.00 ft
Ns (imp. eye flow) / Nss (imp. eye flow)	: 580 / 6,123 US Units
MCSF	: 13.12 USgpm
Head, maximum, rated diameter	: 296.0 ft
Head rise to shutoff (bowl / pump)	: 4.62 / 5.92 %
Flow, best eff. point (bowl / pump)	: 145.7 / 133.1 USgpm
Flow ratio, rated / BEP (bowl / pump)	: 68.18 / 74.64 %
Diameter ratio (rated / max)	: 88.75 %
Head ratio (rated dia / max dia)	: 71.54 %
Cq/Ch/Ce/Cn [ANSI/HI 9.6.7-2010]	: 1.00 / 1.00 / 0.95 / 1.00
Selection status	: Acceptable

Liquid

Liquid type	: Molten Sulphur
Additional liquid description	:
Solids diameter, max	: 0.00 in
Solids concentration, by volume	: 0.00 %
Temperature, max	: 300.0 deg F
Fluid density, rated / max	: 1.790 / 1.790 SG
Viscosity, rated	: 10.00 cP
Vapor pressure, rated	: 0.00 psi.a

Material

Material selected	: Standard
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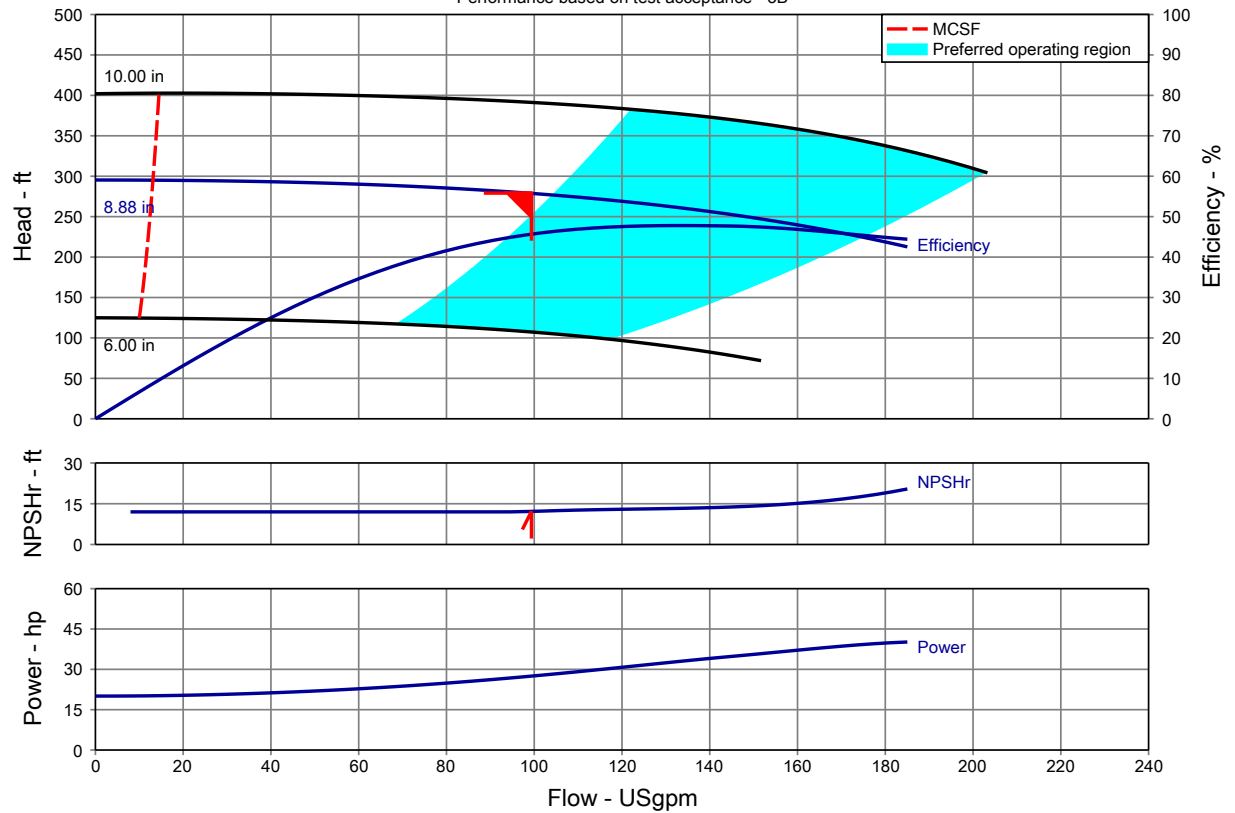
Pressure Data

Maximum working pressure	: See the Additional Data page
Maximum allowable working pressure	: See the Additional Data page
Maximum allowable suction pressure	: N/A
Hydrostatic test pressure	: See the Additional Data page

Driver & Power Data (@Max density)

Driver sizing specification	: Maximum power
Margin over specification	: 0.00 %
Service factor	: 1.00
Power, hydraulic	: 12.71 hp
Power (bowl / pump)	: 27.48 / 27.48 hp
Power, maximum, rated diameter	: 40.17 hp
Minimum recommended motor rating	: 50.00 hp / 37.29 kW

Pump performance. Adjusted for construction, viscosity, static lift to discharge nozzle centerline, friction and power losses of lineshaft and thrust bearings.
The duty point represents the pump performance head.
Performance based on test acceptance - 3B



		SULFUR FEED PUMPS		Revision	
		Doc No. 1510-ME-DTS-4165-EXP-5002		1	

Item	Rev	Client:	Lithium Americas Corp.			
		Project:	Thacker Pass Project		Project No.:	HOU-23001048-A1
					Date:	2024-07-01

1		GENERAL INFORMATION				
2		Equipment Name:	Sulfur Feed Pumps	Vendor:	Weir Pump & Valve Solutions	
3		Number Required	Two (2) - (One is installed spare)	Vendor Quotation Number:	3000280903	
4	1	Equipment Number(s):	1510-PP-004/005	Manufacturer:	WPVS	
5		Specification Number:	1500-ME-SPC-4165-EXP-5001	Model Number:	VSHR-1672	
6		Package Number:	P54165			
7		Instructions to Vendors:				
8		1. Vendors must complete Vendor Data column for each item, otherwise bid may not be considered. "VTA" denotes items for "Vendor to advise" as part of the proposal.				
9		Description	Units	Project Requirement	Vendor Information	
10		SITE CONDITIONS				
11		Site Geographic, Climatic and Seismic Data		Refer to 1000-PE-SPC-0000-BEC-0001		
12		Location (Indoor / Outdoor)		Outdoor		
13		Heated / Unheated		Heated		
14		Hazardous Area Classification		Class 1, Div 2, Group D, Temp Code T3		
15		PROCESS CONDITIONS				
16		Process Description:				
17						
18		Sulfur Feed Pumps pump liquid sulfur from sulfur feed pit to the sulfur burner.				
19						
20		PERFORMANCE				
21		Manufacturer				
22		Pump Type		Glandless, Vertical Spindle submerged	Glandless, Vertical Spindle submerged	
23		Pump Type (cont)		Centrifugal, Steam Jacketed	Centrifugal, Steam Jacketed	
24		Form		VTA	Vertical Submersible	
25		Model No.		VTA	VSHR-1672	
26		Material Spec No.		VTA	105	
27		Operating Flow (per pump) (Note 3)	gpm	75	75	
28		Design Factor		1.15	1.15	
29		Design Flow (per pump) (Note 3)	gpm	86.25	99.36 (Total 86.25 Process + 13.11 mcsf)	
30		NPSH Available	ft	19.34	19	
31		NPSH Required	ft	VTA	12	
32		Design TDH	ft	279.1	279	
33		Efficiency	%	VTA	44.56	
34		BHP, Rating / Maximum	kW	VTA	28.17 / 28.17	
35		Impeller, Max / Design / Min	mm	VTA	10.00 / 8.88 / 6.00	
36		Speed	rpm	VTA	3500	
37		Rotation (viewed from clpg end)		VTA	CCW	
38		Pump Curve Number		VTA	2-2643-62E	
39		Minimum Submergence	ft-in	VTA	1'-3"	
40		Pit Depth from from Bottom of Mounting Flange to pit bottom	ft-in	14'-11"	14'-11"	
42		Utilize standard "J" lengths in quotation. (Note 2)	ft-in	VTA	14'-6"	
42		Suction Inlet	in	VTA / 150	N/A	
43		Discharge Connection Sizes	in	VTA / 150#	2" x 3" / 300# Raised Face	
44						
45		OPERATING CONDITIONS				
46		Fluid Description		Liquid Sulfur	Liquid Sulfur	
47		Liquid Temperature	°F	260 - 300	260 - 300	
48		Liquid Specific Gravity, (Normal/Maximum)		1.79	1.79	
49		Liquid Viscosity, (Normal / Maximum)	cP	6.6 - 10.0	6.6 - 10.0	
50		Sump Pressure	psia	12.3	12.3	
51		Duty		Continuous	Continuous	
52		Sulfur Liquid Level from bottom of Pit				
53		Low Level	ft-in	4'-0"	4'-0"	
54		Normal	ft-in	6'-10 7/8"	6'-10 7/8"	
55		High	ft-in	11'-0"	11'-0"	
56		DESIGN DETAILS				
57		Design Life		25 years		
58		Motor		See 1500-EL-DTS-3121-EXP-5001		

		SULFUR FEED PUMPS		Revision	
		Doc No. 1510-ME-DTS-4165-EXP-5002		1	

Item	Rev	Client:	Lithium Americas Corp.	Project No.:	HOU-23001048-A1	Date:	2024-07-01
		Project:	Thacker Pass Project				
59		Manufacturer		VTA		WEG	
60		Frame Size		VTA		326TSD	
61		Horsepower Rating	HP	VTA		50	
62		Speed	rpm	1800		1800	
63		Coupling		VTA		Thomas DBZ	
64		Manufacturer		VTA		Thomas DBZ	
65		Type		Spacer		Spacer	
66		Coupling Guard		OSHA approved		OSHA approved	
67		Shaft Grounding Brush Kit		VTA		N/A	
68							
69		MATERIALS					
70		Impeller		VTA		Cast Iron	
71		Casing		VTA		Cast Iron	
72		Shaft		VTA		Carbon Steel	
73		Mounting Plate		VTA		Carbon Steel	
74		Column		VTA		Carbon Steel	
75		Casing Gasket		VTA		Non-Asbestos Comp.	
76		Impeller O-ring		VTA		N/A	
77							
78		ELECTRICAL					
79		Motor		Vendor to Provide Motor		Vendor to Provide Motor	
80		Requirements		1500-EL-DTS-3121-EXP-5001_0 - LV (460V) Induction Motors Data Sheet		1500-EL-DTS-3121-EXP-5001_0 - LV (460V) Induction Motors Data Sheet	
81		Manufacturer		WEG		WEG	
82		Voltage/Phase/Frequency	V/ph/Hz	460/3/60		460/3/60	
83		VFD Duty Required		Not Required		Not Required	
84							
85		ANCILLARY COMPONENTS					
86		Coupling					
87		Manufacturer		VTA		Thomas DBZ	
88		Type		flexible type		flexible type	
89		Coupling Guard		Aluminum (Non-sparking)		Aluminum (Non-sparking)	
90		Mounting Plate Size		VTA		32" x 22"	
91		Seal Type		VTA		1/4" Graphited PTFE Packing	
92		Seal Flush Plan		VTA		N/A	
93		Shaft Grounding Brush Kit		Required		N/A	
94		Pump Bearing Lubrication		VTA		Grease	
95		Cooling Requirement		VTA		N/A	
96		Bearing House Vibration Sensors		Yes		No	
97		Pump Bearing RTD		Yes		No	
98		Motor Amps Meter		Yes		No	
99		Junction Box		Yes		No	
100							
101		SURFACE TREATMENT					
102		Type of Coating		Per 1000-ME-SPC-0000-BEC-0024		MFG. Standard	
103		Coating Dry Film Thickness		Per 1000-ME-SPC-0000-BEC-0024		MFG. Standard	
104		No. of coats		Per 1000-ME-SPC-0000-BEC-0024		MFG. Standard	
105							
106		SHOP TESTING & INSPECTION					
107		Shop Hydrotest		Per 1500-ME-SPC-4165-EXP-5001		77068.ITP	
108		Shop Performance Test		Per 1500-ME-SPC-4165-EXP-5001		77068.ITP	
109		Alignment of Mechanical Components		VTA		N/A	
110		Overheating of Mechanical Components		VTA		Included w/Performance Test	
111		Open Tank Test		VTA		N/A	
112		Excess Vibration		VTA		Included w/Performance Test	
113		Noise Levels		VTA		Included w/Performance Test	
114		Clearances		VTA		N/A	
115							
116		SHIPPING DATA					
117		Project Shipping Restrictions (l x w x h)	ft	50 x 24 x 16			
118		Overall Equipment Assembled Sizing (l x w x h)	ft	VTA		TBD (TIME OF SHIPMENT)	

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Item	Rev	Client: Lithium Americas Corp. Project: Thacker Pass Project		Project No.:	HOU-23001048-A1	Date:	2024-07-01
119		Number of Shipping Pieces		VTA	TBD (TIME OF SHIPMENT)		
120		Total Shipping Weight		lb	VTA	TBD (TIME OF SHIPMENT)	
121		Operating Weight		lb	VTA	TBD (TIME OF SHIPMENT)	
122		Full Weight		lb	VTA	TBD (TIME OF SHIPMENT)	
123		Total Shipping Volume		f ³	VTA	TBD (TIME OF SHIPMENT)	
124		Overweight Shipping Required?		VTA	TBD (TIME OF SHIPMENT)		
125		Heaviest Component for Maintenance		item, lb	VTA	TBD (TIME OF SHIPMENT)	
126		Heaviest Component for Shipping		item, lb	VTA	TBD (TIME OF SHIPMENT)	
127		Oversized Shipping Required?		VTA	TBD (TIME OF SHIPMENT)		
128		Largest Shipping Piece, item (l x w x h)		ft	VTA	TBD (TIME OF SHIPMENT)	
129		Number of Components for Site Erection		VTA	TBD (TIME OF SHIPMENT)		
130		FOB Point(s)		VTA	TBD (TIME OF SHIPMENT)		
131		Note(s)					
132							
133		1. Pump motor to be sized for maximum power absorbed on curve, plus 15%.					
134		2. If an extension is required on pump suction, it shall be steam jacketed.					
135		3. Vendor to drill recycle hole in the discharge of each pump to protect against closed downstream valve. Vendor to account for the					
136		additional recycle flow at the design point.					
137		VENDOR COMMENTS:					
138		9аааУІ-іІІІ LІІІYU 6SІІKІY мулн f0ā					
139							
140							
141							
142							

Utility Consumption List

Certified for: LITHIUM NEVADA LLC			Weir Minerals Lewis Pumps 8625 Grant Rd St. Louis, MO 63123 Telephone 314-843-4437 sales.lewis@weirminerals.com 	
Customer: LITHIUM NEVADA LLC				
PO: 4500001187				
Tag #s: 1510-PP-001/002, 1510-PP-004/005				
Rev	Date	Description	Document #:	77067-68.UTL
0	August 20, 2025	Original issue	Prepared by:	Mason Seidel
			Approved by:	R. Chickering
			Serial #s:	77067-68
			Pump Type:	MSS-1655, VSHR-1682
			Page:	1 of 1

Sound Pressure Level (dBA)

Pump Design: <85 dBA Load Measurement: 100%